STATIC TIMING ANALYSIS REPORT

OpenSTA Multi-Corner PVT Analysis

Source: User-supplied STA result table and terminal execution screenshots

1. Executive Summary

The supplied OpenSTA results contain **16 PVT test cases** covering FF, SS and TT process conditions. Based on the provided table, **8 cases are marked MET** and **8 cases are marked VIOLATED**. The terminal screenshots also show individual STA runs completing successfully and a final **STA RUN COMPLETE** message.

The largest positive maximum slack is **5.59 ns** at FF / -40°C / 1.95 V. The most negative maximum slack is **-62.56 ns** at SS / -40°C / 1.28 V. All supplied minimum-slack values are positive, ranging from **0.19 ns** to **1.83 ns**.

Metric	Result
Total PVT cases	16
MET cases	8
VIOLATED cases	8
Worst Max Slack	-62.56 ns — SS / -40°C / 1.28 V
Best Max Slack	5.59 ns — FF / -40°C / 1.95 V
Min Slack range	0.19 to 1.83 ns

2. Analysis Method

The analysis treats each row in the supplied table as one PVT timing-analysis condition. The Process, Temperature and Voltage columns identify the corner, while the reported minimum and maximum slack values are retained exactly as supplied. The plots below visualize the same 16 rows without aggregation of the individual test cases.

Note: The report uses the table's column labels exactly as provided. A negative value in the supplied Max Slack column is therefore interpreted as a timing violation because the Status column also marks those rows as VIOLATED.

3. Detailed PVT Results

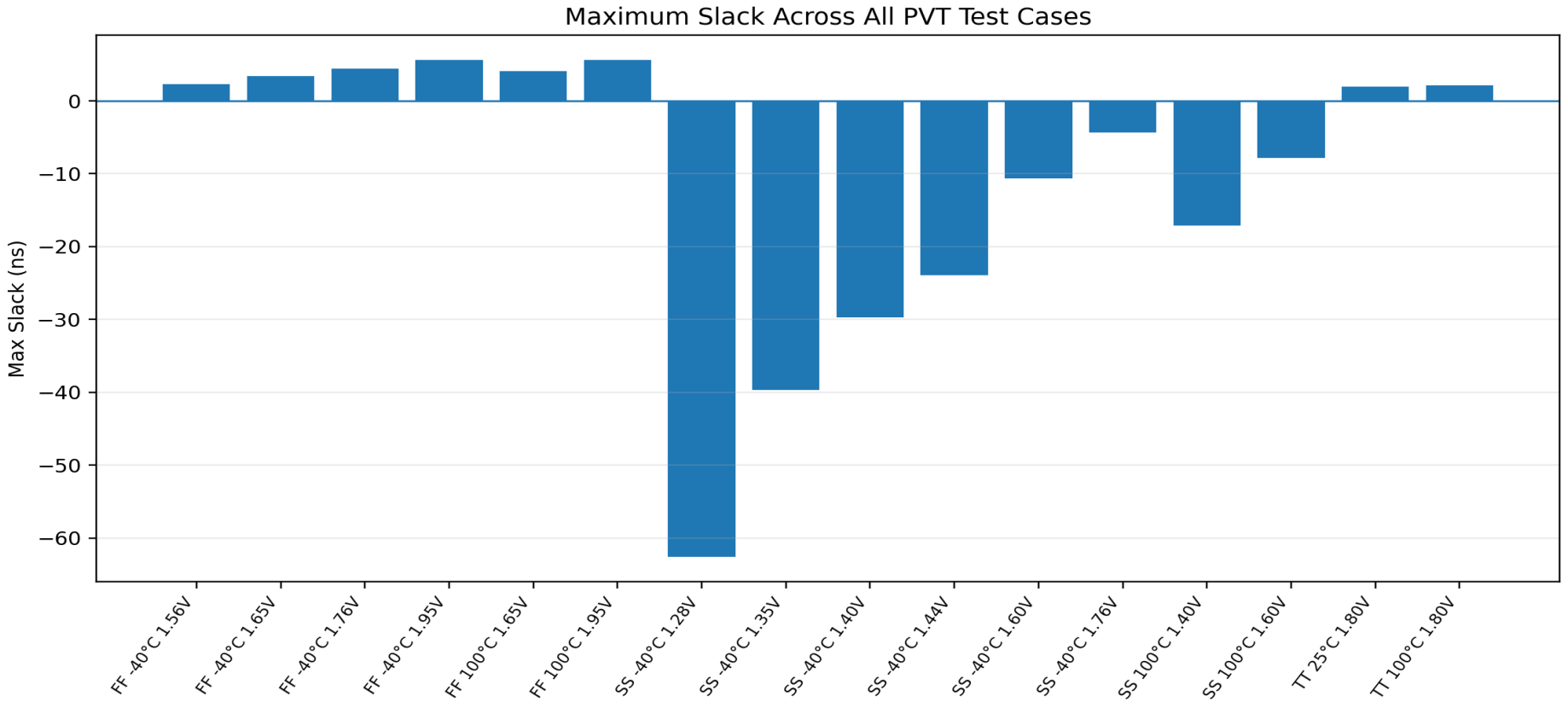
#	Process	Temp	Voltage	Min Slack (ns)	Max Slack (ns)	Status
1	FF	-40°C	1.56	0.29	2.28	MET
2	FF	-40°C	1.65	0.26	3.41	MET
3	FF	-40°C	1.76	0.22	4.43	MET
4	FF	-40°C	1.95	0.19	5.59	MET
5	FF	100°C	1.65	0.25	4.05	MET
6	FF	100°C	1.95	0.20	5.57	MET
7	SS	-40°C	1.28	1.83	-62.56	VIOLATED
8	SS	-40°C	1.35	1.35	-39.70	VIOLATED
9	SS	-40°C	1.40	1.12	-29.69	VIOLATED
10	SS	-40°C	1.44	0.99	-23.91	VIOLATED
11	SS	-40°C	1.60	0.66	-10.61	VIOLATED
12	SS	-40°C	1.76	0.50	-4.38	VIOLATED
13	SS	100°C	1.40	0.91	-17.12	VIOLATED
14	SS	100°C	1.60	0.64	-7.87	VIOLATED
15	TT	25°C	1.80	0.31	1.94	MET
16	TT	100°C	1.80	0.31	2.09	MET

Key observations

- FF conditions are marked MET for all six supplied FF cases.
- SS conditions are marked VIOLATED for all eight supplied SS cases; their maximum slack values are negative.
- TT conditions are marked MET for both supplied TT cases.
- The minimum-slack column remains positive for every row in the supplied table.

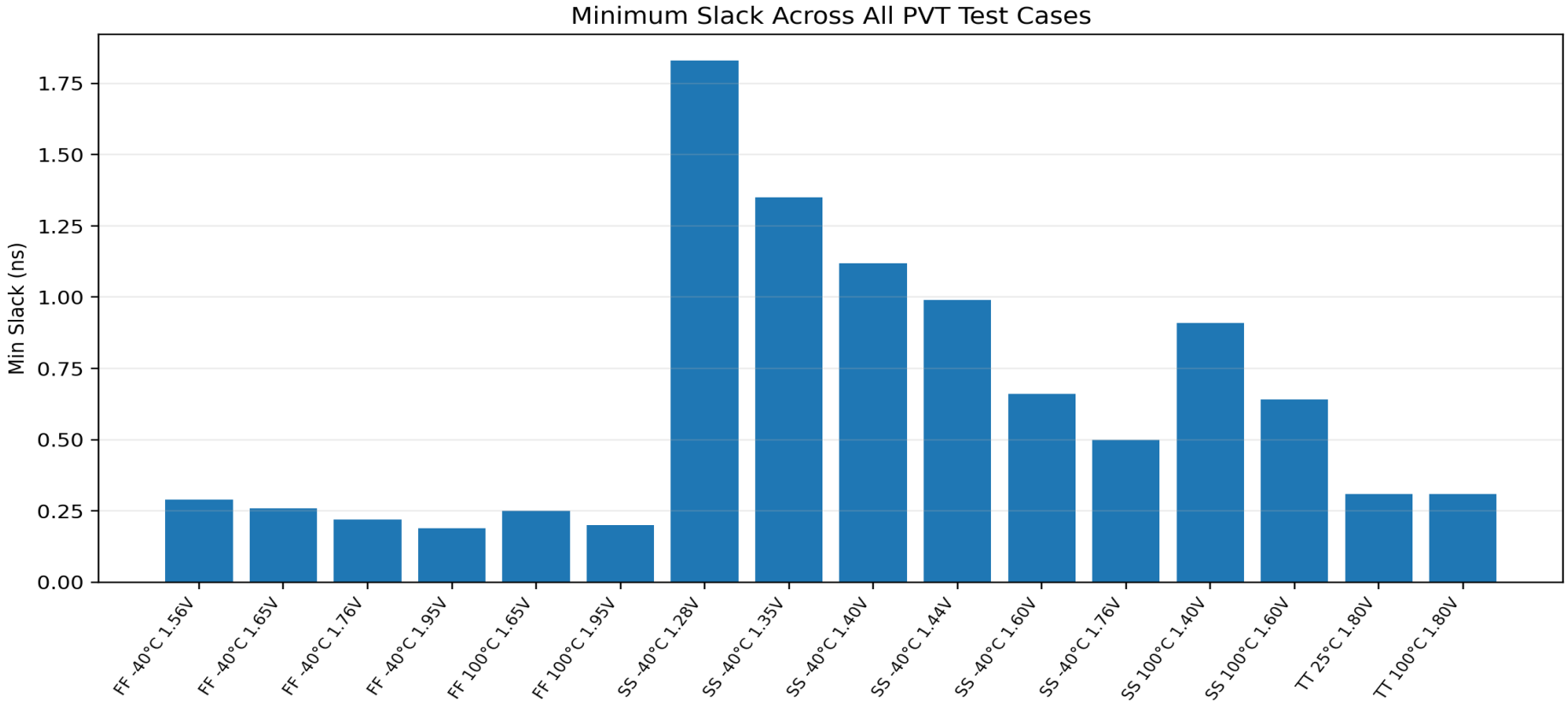
4. Maximum Slack Trend

The following chart compares the supplied Max Slack value for every PVT test case. The zero reference separates positive slack from negative slack; the SS cases occupy the negative region.



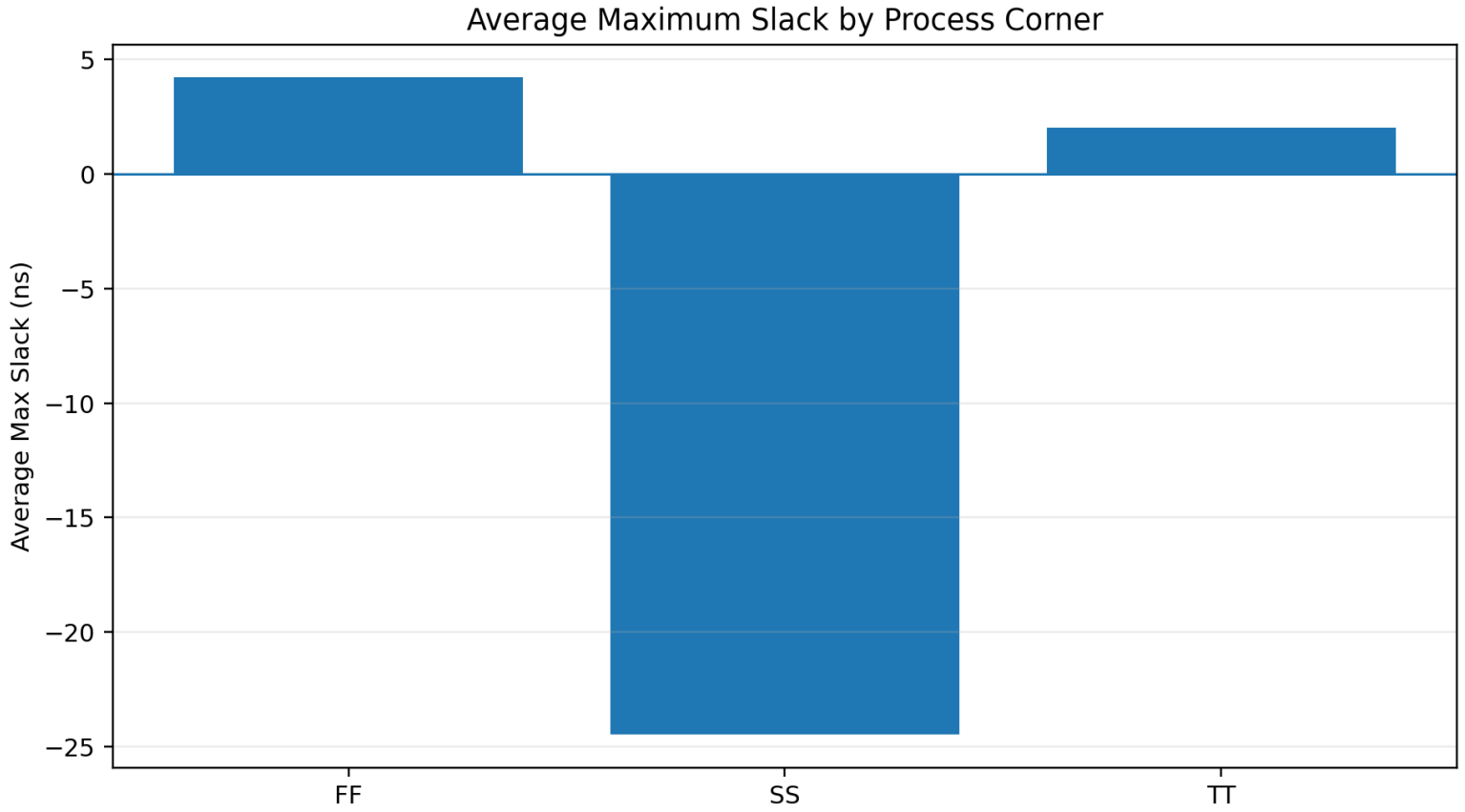
5. Minimum Slack Trend

All 16 supplied minimum-slack values are above zero. The chart makes the relative margin across the PVT cases visible.



6. Process-Level Summary

Averaging the supplied Max Slack values within each process group shows a clear separation between FF/TT and SS in this dataset. This is a descriptive summary of the provided results, not a replacement for detailed path-level timing analysis.



Process	Number of cases	Average Max Slack (ns)	Range of Max Slack (ns)
FF	6	4.22	2.28 to 5.59
SS	8	-24.48	-62.56 to -4.38
TT	2	2.01	1.94 to 2.09

7. STA Execution Evidence

The supplied screenshots show the OpenSTA workflow running the PVT cases, producing result files, reporting slack/status values, and completing the STA run. The images are included as evidence for the execution flow.

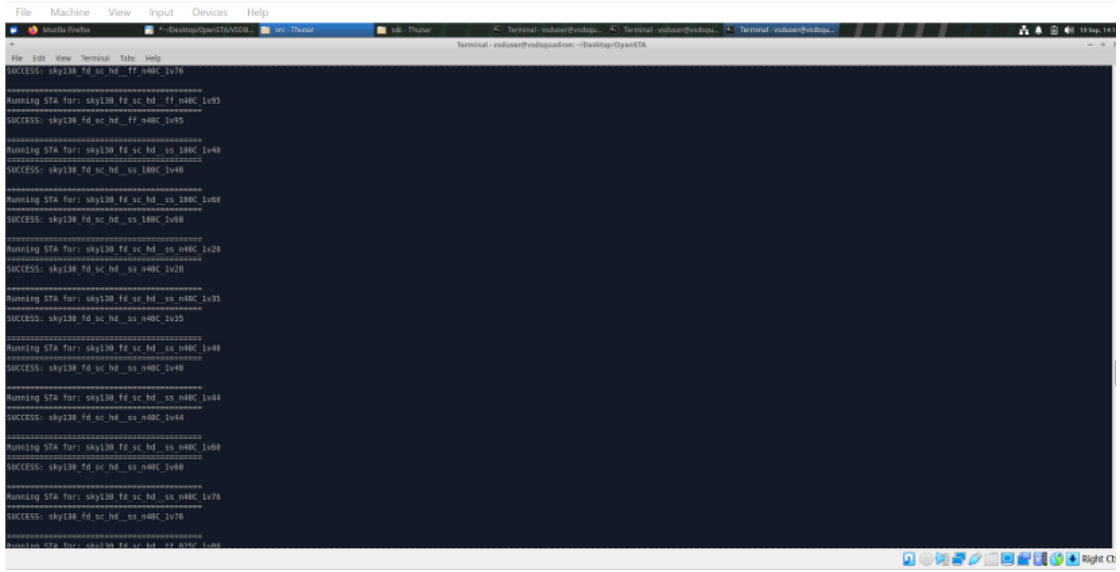
Evidence image 1

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3	FF	-40°C	1.76 V	0.22	4.43	MET
4	FF	-40°C	1.95 V	0.19	5.59	MET
5	FF	100°C	1.65 V	0.25	4.05	MET
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7	SS	-40°C	1.28 V	1.83	-62.56	VIOLATED
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9	SS	-40°C	1.40 V	1.12	-29.69	VIOLATED
10	SS	-40°C	1.44 V	0.99	-23.91	VIOLATED
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12	SS	-40°C	1.76 V	0.50	-4.38	VIOLATED
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16	TT	100°C	1.80 V	0.31	2.09	MET

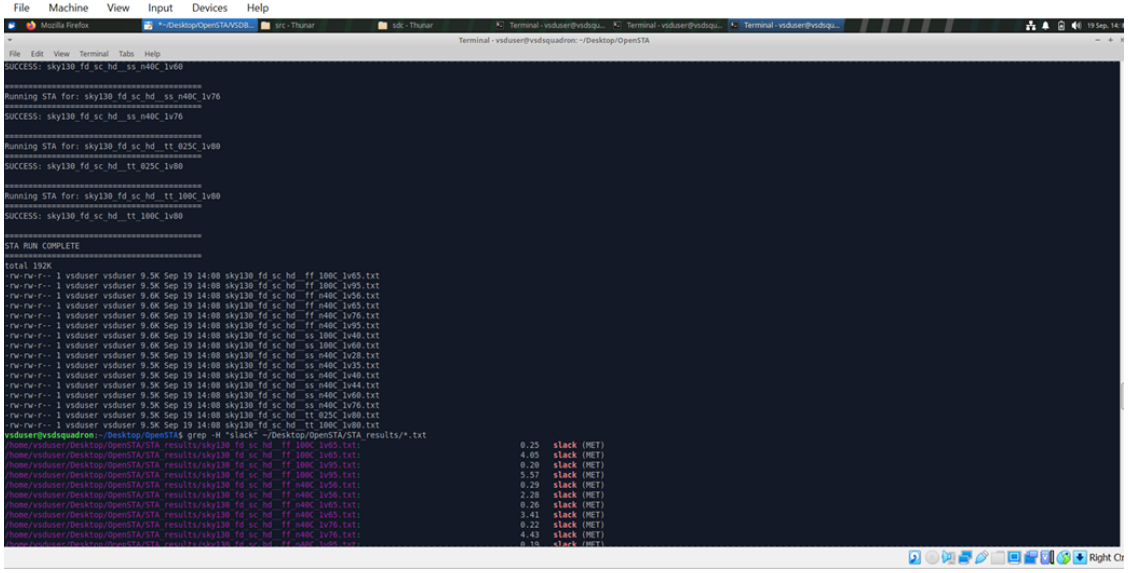
Evidence image 2


File Machine View Input Devices Help
Terminal - vsduser@vsdsquadron: ~/Desktop/OpenSTA
File Edit View Terminal Tabs Help
OpenSTA 3.1.0 1b9a78e8c4 Copyright (c) 2026, Parallax Software, Inc.
License GPLv3: GNU GPL version 3 <http://gnu.org/licenses/gpl.html>
This is free software, and you are free to change and redistribute it
under certain conditions; type `show copying' for details.
This program comes with ABSOLUTELY NO WARRANTY; for details type `show warranty'.
Error 340: read_liberty line 1, cannot open 'read_liberty'.
vsduser@vsdsquadron:~/Desktop/OpenSTA$ rm -f ~/Desktop/OpenSTA/STA_results/*.txt
vsduser@vsdsquadron:~/Desktop/OpenSTA$ nano run_all_corners.sh
vsduser@vsdsquadron:~/Desktop/OpenSTA$ nano run_all_corners.sh
vsduser@vsdsquadron:~/Desktop/OpenSTA$ chmod +x run_all_corners.sh
vsduser@vsdsquadron:~/Desktop/OpenSTA$ ./run_all_corners.sh
Running STA for: sky130_fd_sc_hd__ff_100C_1v65
SUCCESS: sky130_fd_sc_hd__ff_100C_1v65
Running STA for: sky130_fd_sc_hd__ff_100C_1v95
SUCCESS: sky130_fd_sc_hd__ff_100C_1v95
Running STA for: sky130_fd_sc_hd__ff_n40C_1v56
SUCCESS: sky130_fd_sc_hd__ff_n40C_1v56
Running STA for: sky130_fd_sc_hd__ff_n40C_1v65
SUCCESS: sky130_fd_sc_hd__ff_n40C_1v65
Running STA for: sky130_fd_sc_hd__ff_n40C_1v76
SUCCESS: sky130_fd_sc_hd__ff_n40C_1v76
Running STA for: sky130_fd_sc_hd__ff_n40C_1v95
SUCCESS: sky130_fd_sc_hd__ff_n40C_1v95
Running STA for: sky130_fd_sc_hd__ss_100C_1v40
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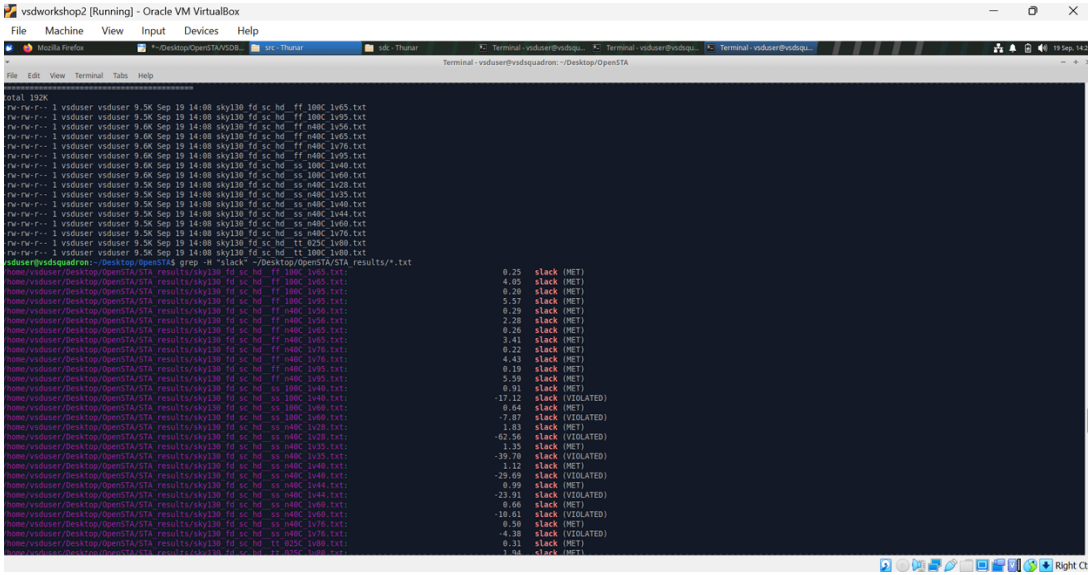
Evidence image 3



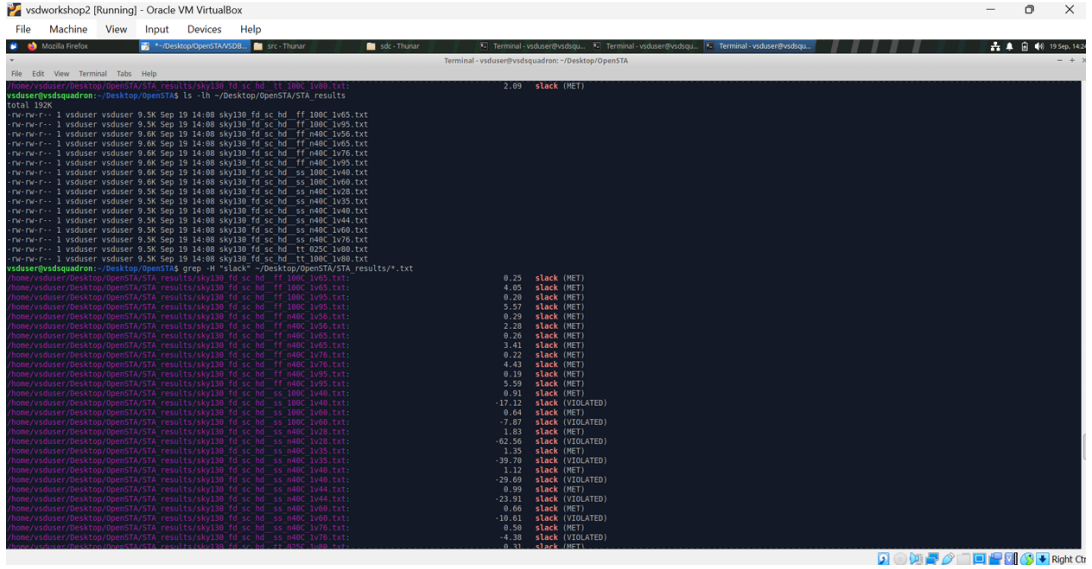
Evidence image 4



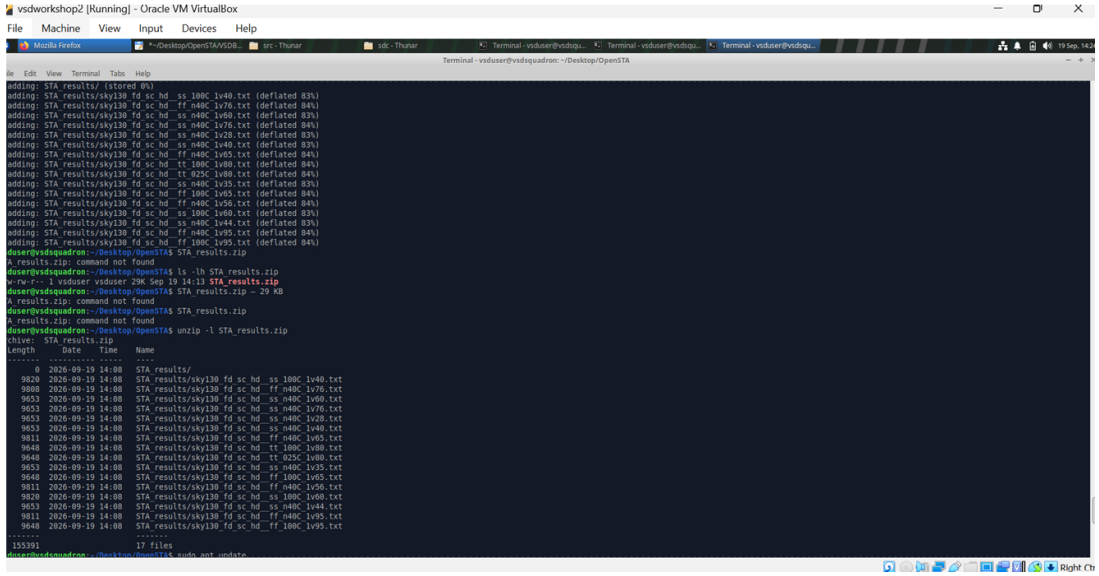
Evidence image 5



Evidence image 6



Evidence image 7



8. Conclusion

The supplied OpenSTA table contains 16 PVT timing cases, with 8 marked MET and 8 marked VIOLATED. The critical result in the table is the SS / -40°C / 1.28 V case, which has a Max Slack of -62.56 ns, while all supplied FF and TT cases are marked MET. The terminal evidence indicates that the STA scripts completed and generated the corresponding result files.

> **Recommended next step:** Use the violated SS corners to identify the critical timing paths and investigate whether cell sizing, buffering, logic restructuring, clock constraints, or other timing-closure techniques are appropriate for the design.

End of Report